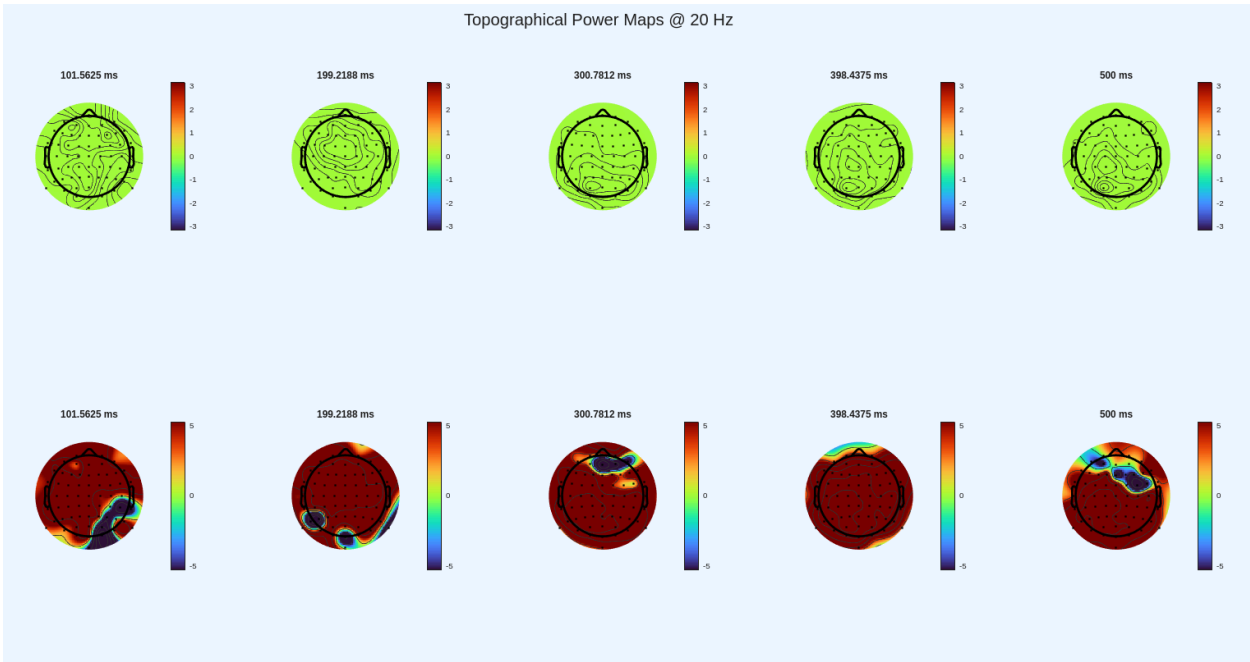
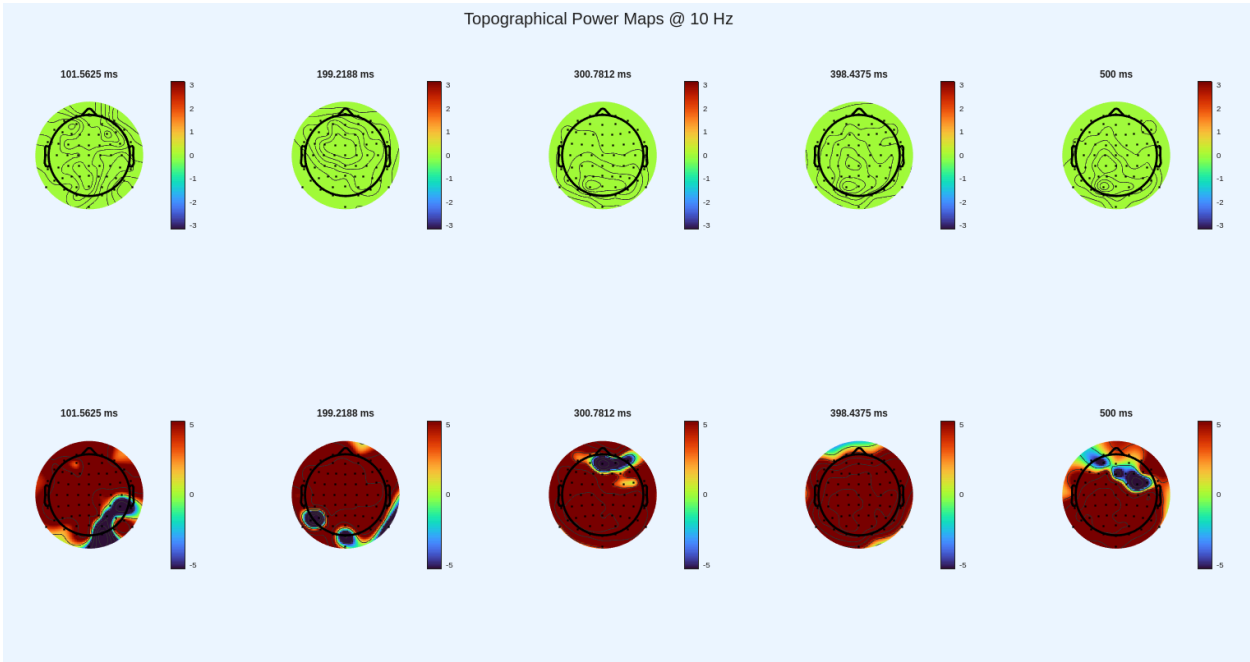
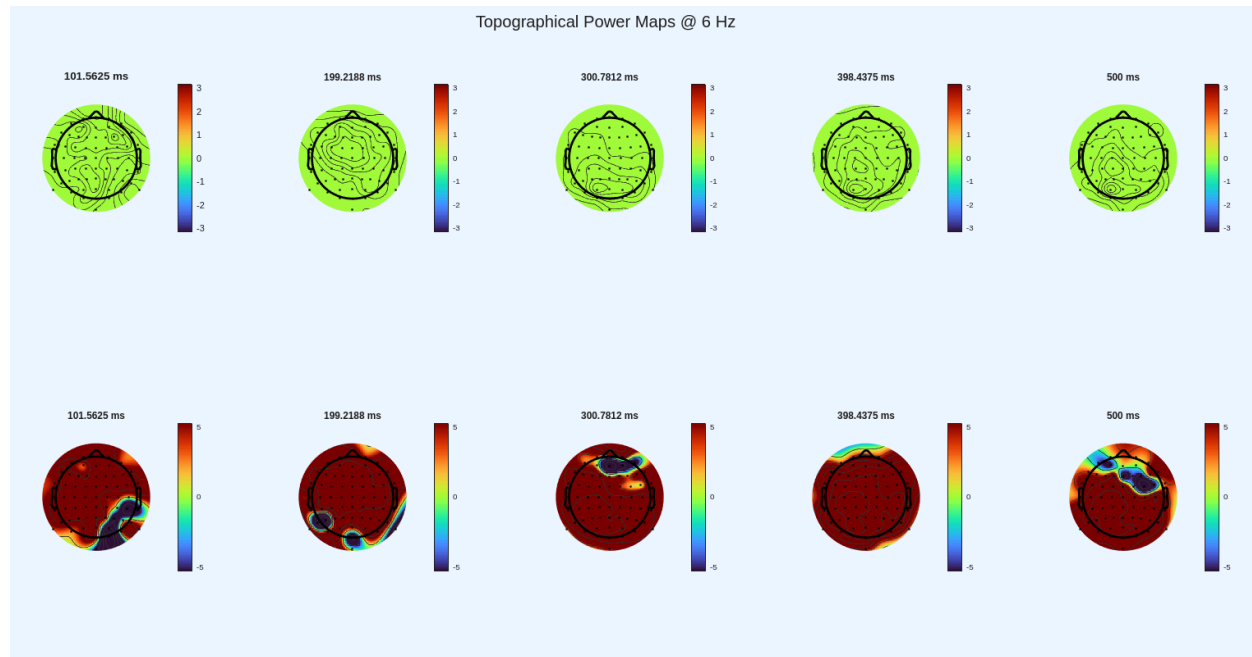


CHAPTER 18 RESULTS AND INTERPRETATION





Results and Interpretation

When comparing the topographical power maps with and without baseline normalization, clear differences can be seen. The plots without baseline correction mainly show overall power levels across the scalp, and they look fairly uniform over time. After baseline normalization, the maps highlight changes relative to the pre-stimulus period — regions of increased or decreased power become much more visible.

The differences are especially clear in the lower frequency bands. At **6 Hz (theta)**, there is stronger activity over frontal and central areas after around 200 ms, which likely reflects task-related theta synchronization. At **10 Hz (alpha)**, there is a clear decrease in power over occipital electrodes, showing typical alpha desynchronization after stimulus onset. The **20 Hz (beta)** maps show smaller and more localized effects, mostly over central regions.

Overall, baseline normalization makes it easier to see real event-related changes because it removes general channel-to-channel differences in power. Without it, the maps mostly reflect absolute signal strength rather than functional modulation. The differences are most prominent in theta and alpha bands and around 200–500 ms post-stimulus. These changes happen because baseline correction expresses power as a relative change from pre-stimulus activity, which emphasizes event-related synchronization (ERS) and desynchronization (ERD) patterns across the scalp.